Evolutionary Factor Research:
Baseline Results and Ablations

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Abstract

All runs share one protocol on the survivorship-bias-free Nasdaq-100 point-in-time panel (2010–2026): factors are researched on data up to mid-2021, and the five years 2021–2026 are consumed as ten six-month blocks, each scored *before* the researcher may adapt to it — an honest walk-forward record. Part 1 examines the full evolutionary configuration (knowledge-graph-grounded ideation, four-objective selection, progressive data reveal) in detail; Part 2 removes one ingredient at a time to locate where the out-of-sample value actually comes from. All numbers are signal-level information coefficients (IC); no position construction or costs enter.

1 The baseline: full evolutionary configuration

Twenty generations, eight mechanism groups derived from a knowledge graph of 1,723 papers, three demes per group, LLM mutation/crossover operators, and a four-objective Pareto selection (marginal value to the book, independence, parsimony, structural novelty). The run produced 57 factors for $250 in LLM cost across roughly 900 scored candidates.

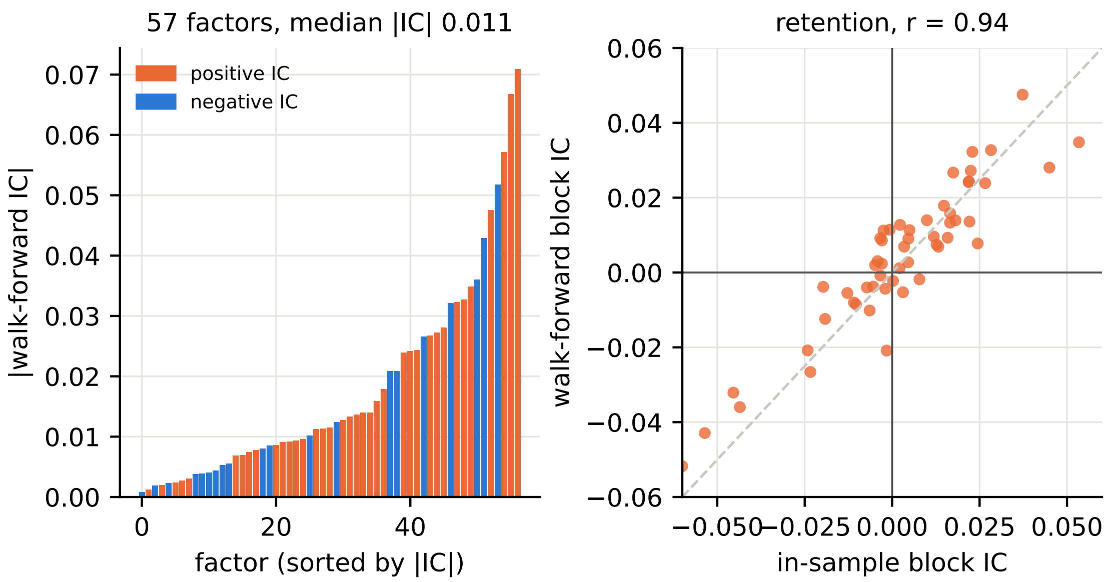

Figure 1: Left: the 57 surviving factors' absolute walk-forward ICs (|mean over the ten held-out blocks| — a stably negative factor is just as predictive as a positive one and is sign-flipped in combination); median |IC| 0.011, 65% positive-signed. Right: in-sample versus walk-forward signed IC per factor. Per-factor edge is essentially fully retained out of sample (r = 0.94; only 18% of factors flip sign; mean |IC| 0.019 in-sample vs. 0.018 walk-forward).

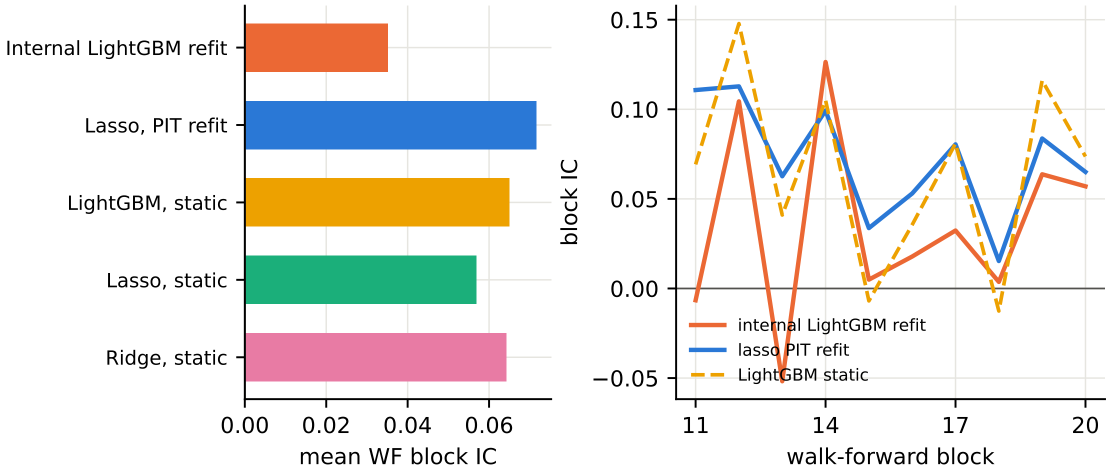

Figure 2: Combining the book into one signal. The prequential LightGBM refit the search itself uses (orange) averages 0.035; a point-in-time lasso refit per block reaches 0.072 and a static ridge 0.064 — the factor set is better than the score the evolution sees. The lasso is positive in every block.

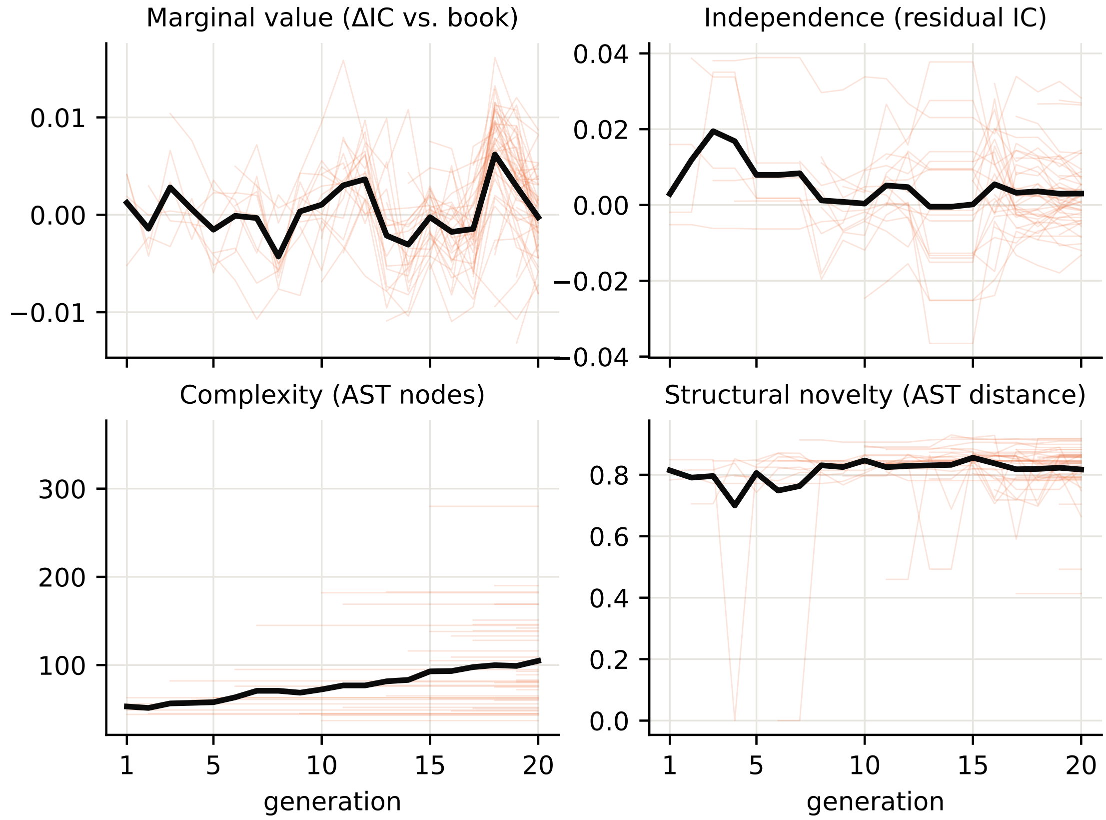

Figure 3: The four selection objectives across the final archive over the evolution (thin lines: individual factors, carried forward between re-scores; black: archive mean). Marginal value fluctuates around zero by construction (each survivor is measured against the rest of the book) and independence compresses toward zero as the book fills its niches; complexity roughly doubles over twenty generations while structural novelty stays flat at ~0.85 — the parsimony objective slows but does not stop code growth.

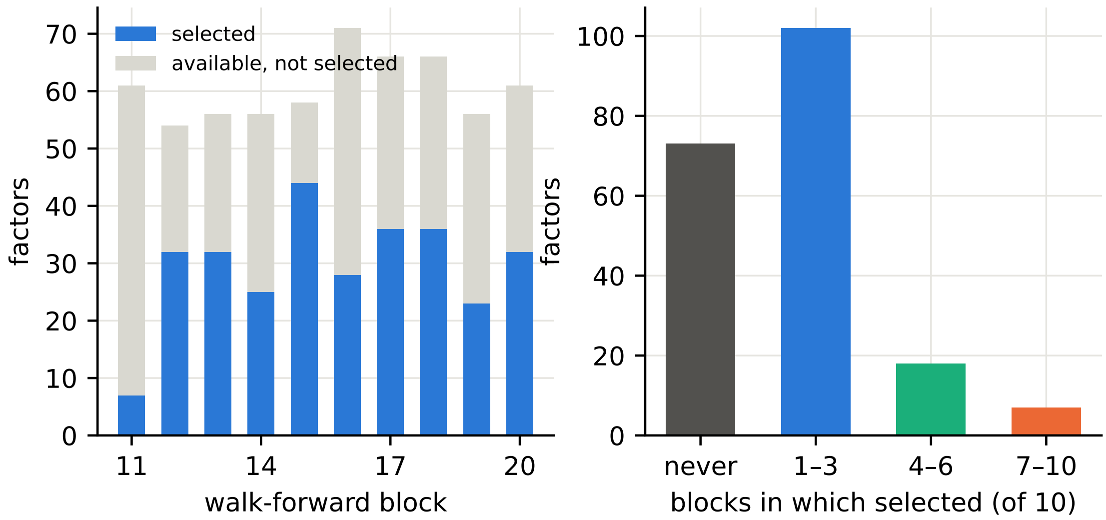

Figure 4: What a sparse combiner actually uses. Per block the cross-validated lasso keeps 7–44 of the ~60 available factors (left); over the ten refits 37% of the ~200 factors that ever entered the point-in-time pool (archive snapshots grow through the walk-forward) are never selected and under 4% are selected in most blocks (right) — yet the combined IC is stable, so the pool is redundant in a robust, interchangeable sense. Diversity confirms this: mean pairwise $|\rho| = 0.09$, effective dimension $\approx$ 22 of 57 by the eigenvalue-participation ratio $N_{\text{eff}} = (\sum \lambda_i)^2 / \sum \lambda_i^2$.

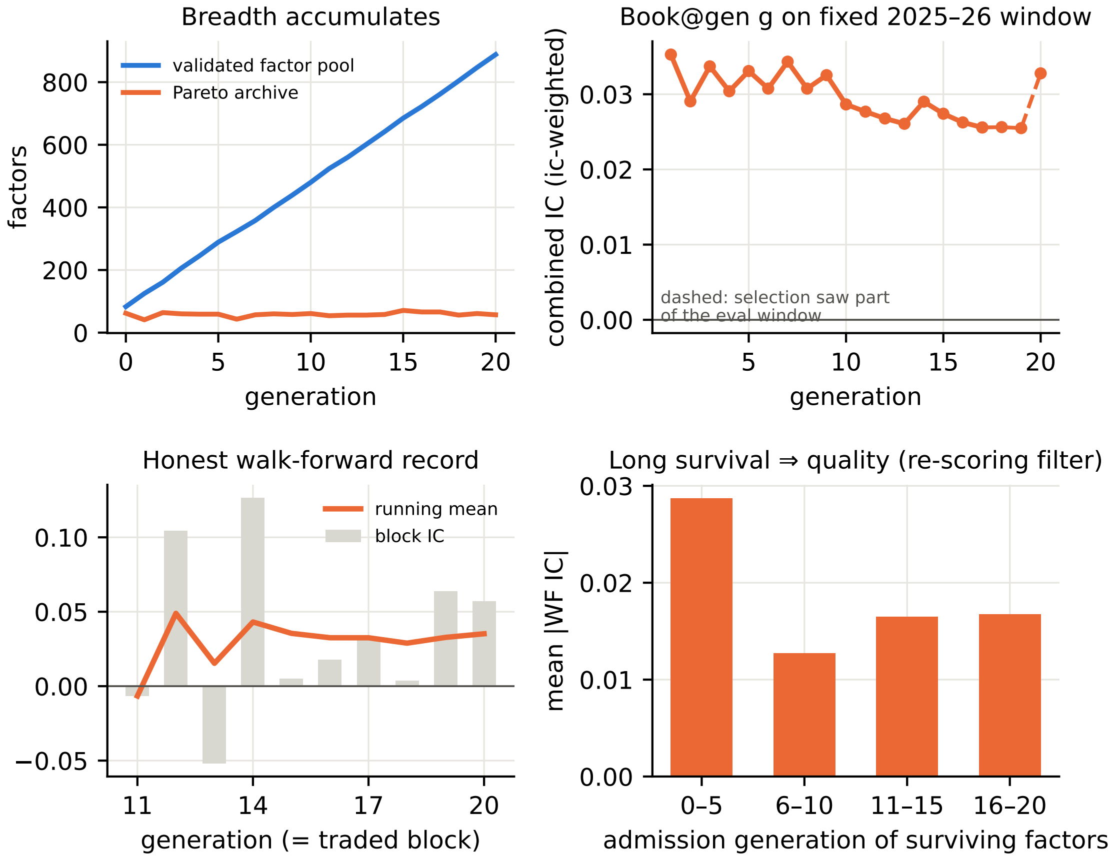

Figure 5: What actually changes over the evolution (baseline arm). Top left: the validated factor pool grows linearly (83 → 890) while the curated archive holds at the cap — evolution accumulates breadth. Top right: the archive *as of each generation*, recombined with identical ic-weights fit before a fixed 2025–26 window and scored on it — the combined IC is flat: the generation-1 book was already as good as any later one, and composition churn adds no combined value on untouched future data. Bottom left: the honest prequential record accumulates and stabilises at $\approx +0.035$ — flatness under 20 generations of selection pressure is the anti-overfitting machinery holding (a naive search would decay; cf. the IC-only arm). Bottom right: surviving factors admitted earliest — those re-scored up to twenty times on growing, partially unseen windows — have twice the realised walk-forward |IC| of later admits: survival through re-scoring, not evolution's churn, is the effective quality filter.

2 Where the value comes from: ablations

Each arm below removes or adds one ingredient relative to the baseline, under the identical panel, split schedule and scoring harness.

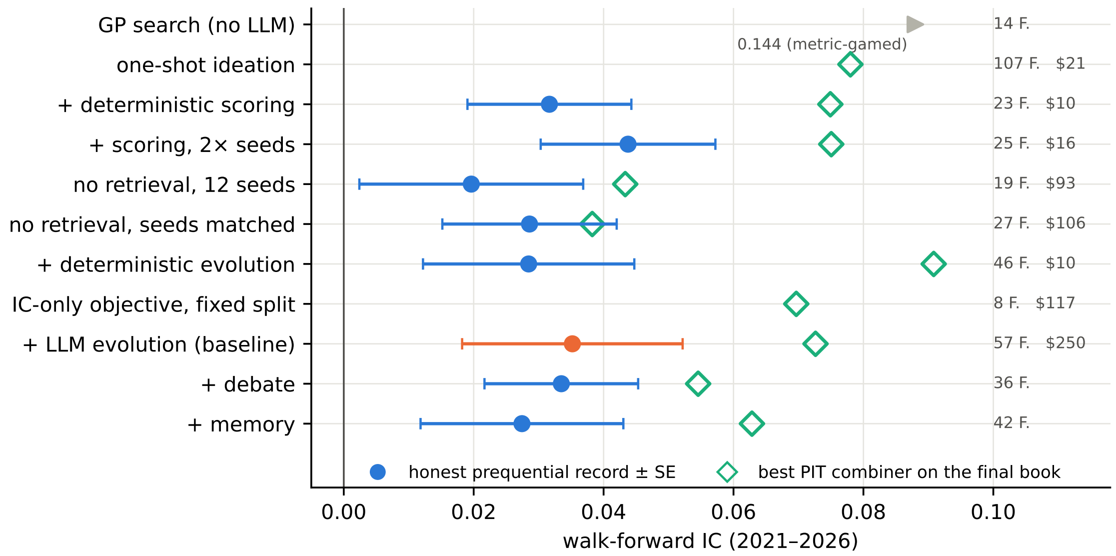

Figure 6: The ablation ladder. Filled dots: the honest prequential record (mean of the ten block ICs ± standard error); open diamonds: the best point-in-time combiner applied to the arm's final book; right margin: book size and LLM cost. The GP arm's record is metric-gamed (see text) and shown clipped.

Three readings. First, *ideation plus deterministic scoring carries almost everything*: seeding ideas once and letting the deterministic harness score, prune and re-curate them (no evolution at all, $10–16) lands within block noise of the full $250 baseline, and doubling the seed budget produced the best honest record of the family (+0.044). Second, *the LLM inside the evolution adds little*: replacing all LLM mutation by deterministic window-jitter (+0.028) or removing children entirely (+0.032) is statistically indistinguishable from the baseline (+0.035, block SE $\approx$ 0.012) — and under the best point-in-time combiner the jitter-evolved book is in fact the family's best (0.091, ten of ten blocks positive, at $10). Third, *the LLM's economic prior is what keeps the search honest*: the prior-free GP miner "won" with +0.144 by discovering degenerate statistics (raw price and fundamental levels whose per-underlying IC telescopes mean reversion), and after a stationarity gate was added it evaded the gate with near-boundary level proxies — a Goodhart demonstration that also explains *why* the semantically grounded arms do not overfit the same way.

Retrieval grounding needed a control to price correctly. The first two no-retrieval runs (+0.020 and +0.012) were accidentally seeded with only 10–12 ideas instead of 96 (an oversized idea-generation call silently timed out), confounding retrieval with seeding breadth. The seeding-matched rerun resolves it: with 96 ungrounded ideas the no-retrieval arm reaches +0.029 — statistically indistinguishable from the baseline's +0.035 on the internal record — yet its factors are individually the weakest of the whole family (median |IC| 0.005 vs. 0.011; 44% flip sign out of sample vs. 18%; effective dimension 14 vs. 22, including one near-duplicate pair), and its book recombines point-in-time to only 0.038 versus the baseline's 0.073. Retrieval grounding therefore does not move the noisy headline record, but roughly doubles the deployable, combined walk-forward IC of the resulting book. Debate and cross-run memory did not improve on the baseline. (One mechanical caveat: in the debate arms a sparse-coverage factor admitted early silently disabled the residual-IC independence objective for nearly all candidates — those two runs effectively selected on three objectives.)

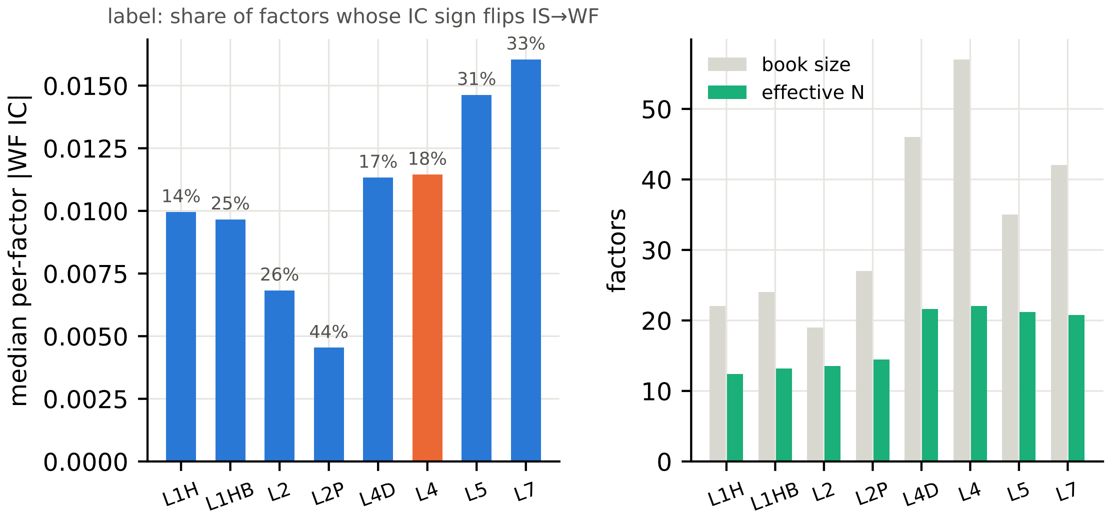

Figure 7: Book quality across arms (baseline in orange). Left: median per-factor absolute walk-forward IC, with the share of factors whose IC *sign flips* from fit window to walk-forward printed above each bar. The debate/memory books post the highest raw |IC| medians — but with the worst sign stability (31–33% flips vs. 18% for the baseline), and a quarter of their members are exact quarterly-stepped (ρ = 1.0) level-like signals whose per-underlying IC is mechanically inflated (9/35 and 11/42 vs. 4/57 in the baseline). Right: book size versus effective independent dimension. The baseline leads where it matters for deployment: sign stability, clean composition, breadth — and the combined point-in-time results.

3 Key observations

1. **The deterministic scoring harness, not the evolutionary loop, is the workhorse.** Ideation + scoring at \$10–16 matches twenty generations of LLM evolution at \$250 on the honest five-year record; trial counts (81 vs. ~900) do not translate into overfitting because the anti-ratchet machinery (progressive reveal, re-scoring, deflation) holds the record flat.

2. **Factor-level edge transfers; the combiner is the fragile part.** Per-factor |IC| is fully retained out of sample in every arm, while the deployment combiner moves the headline number by a factor of two — report combiner-robustness, not one combiner's IC.

3. **The LLM's contribution is the economic prior at ideation.** Removing the LLM entirely (GP) collapses into metric gaming that survives a targeted stationarity gate; removing only retrieval grounding (seeds matched) halves the deployable combined IC of the book (0.038 vs. 0.073) even though the headline record barely moves.

4. **Evolution buys sign stability and breadth, not mean performance** — the lowest sign-flip rate, the cleanest book composition, the largest book and highest effective dimension; relevant for downstream combination and capacity, not for the block-mean record.

5. **The scoring design matters at the factor level, less at the book level.** A final harness ablation (identical configuration, but candidates scored by validation |IC| alone on a fixed three-way split — no multi-objective vector, no progressive reveal; 849 candidates, \$117) shows textbook selection overfitting per factor: the archive collapses to one factor per mechanism group (a single-objective front is a single point), promised validation |IC| (median 0.040) is cut to 0.025 realised, and five of the eight survivors' validation sign disagrees with their own fit window — selection on local noise. Yet the *book* survives: eight nearly orthogonal factors (mean $|\rho|$ = 0.03) recombined linearly post strong 2021–2026 results, because the combiner re-learns the unstable signs and the graph's group structure alone preserves diversity (its full candidate pool, point-in-time

recombined, reaches 0.070 — close to, but below, the baseline's 0.073). The multi-objective vector and progressive reveal are what make *individual factors* trustworthy; diversity structure plus a linear combiner provide a surprising amount of insurance at the book level.